

Raising awareness and promoting inclusive learning using a VR-powered educational tool for teaching ASL

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Abstract

In contemporary education, there is a growing need for innovative tools to better cater to a diverse audience. Virtual Reality (VR) offers immersive visual, auditory, and haptic experiences that enhance both recreational and educational applications. This research introduces a serious game that leverages hand tracking through VR to support early-stage learning of American Sign Language (ASL). The game provides an engaging environment focused on mastering the ASL alphabet (*fingerspelling*) by translating user gestures into an interactive ASL sign game. While currently focused on teaching fingerspelling only, the platform's scalable architecture may enable future expansions to full-word signs and additional sign languages.

Developed in Unity with XR hand tracking plugins, the game emphasizes intuitive interaction. ASL fingerspelling was selected for its static gestures and alignment with standalone headset limitations. Interactive learning is a core aspect of gameplay, and players must sign letters to progress, enabling natural acquisition by progressive through the game. The game was evaluated through a quasi-experimental approach focusing on educational effectiveness and usability. 11 participants responded to multiple pre and post questionnaires, with results revealing positive aspects of the game, but also various usability issues that can lead to improved clarity and knowledge retention.

Beyond language acquisition, promoting sign language through emerging technologies raises awareness of inequality, improves communication skills, and advances quality education (SDG 4). This project can support both hard-of-hearing students and their hearing peers, helping bridge communication gaps, and facilitate the reduction of inequalities (SDG 10).

Keywords: Virtual reality, Hand tracking, Sign language, Serious games, Education